

Netflix Originals Data Analysis: Exploring Trends and Insights

Objective:

The objective of this project is to analyze the Netflix Originals dataset using MySQL queries to extract meaningful insights and trends. By performing various SQL operations such as GROUP BY, HAVING, ORDER BY, LIMIT, JOINS, WINDOW FUNCTIONS, and SUBQUERIES, participants will gain hands-on experience in data analysis and be better prepared for data analyst job roles.

Datasets Description:

1. Netflix Originals: This dataset contains information about Netflix Originals including title, genre ID, runtime, IMDb score, language, premiere date, etc.
2. Genre Details: This dataset contains information about genres such as genre ID and genre name.

Analyses to make:

1. What are the average IMDb scores for each genre of Netflix Originals?

```
SELECT
```

```
    gd.genre_name,
```

```
    ROUND (AVG (no.imdb_score), 2) AS average_imdb_score
```

```
FROM netflix_originals AS no
```

```
INNER JOIN genre_details AS gd
```

```
    ON no.genre_id = gd.genre_id
```

```
GROUP BY gd.genre_name
```

```
ORDER BY average_imdb_score DESC;
```

Output	Result #1
genre_name	average_imdb_score
Horror	6.27
Musical	6.27
Drama	6.27
Comedy	6.27
Action	6.27
Mystery	6.27
Science Fiction	6.27
Thriller	6.27
Satire	6.27
Variety Show	6.27
Rom-Com	6.27
Concert Film	6.27
Biopic	6.27
Historical	6.27

2. Which genres have an average IMDb score higher than 7.5?

SELECT

gd.genre_name,

ROUND (AVG (no.imdb_score), 2) AS average_imdb_score

FROM netflix_originals AS no

INNER JOIN genre_details AS gd

ON no.genre_id = gd.genre_id

GROUP BY gd.genre_name

HAVING AVG (no.imdb_score) > 7.5

ORDER BY average_imdb_score DESC;

Output	Result #1
*	

3. List Netflix Original titles in descending order of their IMDb scores.

```
SELECT
    title,
    imdb_score
FROM netflix_originals
ORDER BY imdb_score DESC;
```

title	imdb_score
David Attenborough: A Life on Our Planet	9.0
Emicida: AmarElo - It's All For Yesterday	8.6
Springsteen on Broadway	8.5
Ben Platt: Live from Radio City Music Hall	8.4
Taylor Swift: Reputation Stadium Tour	8.4
Winter on Fire: Ukraine's Fight for Freedom	8.4
Cuba and the Cameraman	8.3
Dancing with the Birds	8.3
13th	8.2
Disclosure: Trans Lives on Screen	8.2

4. Retrieve the top 10 longest Netflix Originals by runtime.

```
SELECT
    title,
    runtime
FROM netflix_originals
ORDER BY runtime DESC
LIMIT 10;
```

Output	Result #1	
title	runtime	
The Irishman	209	
Da 5 Bloods	155	
Springsteen on Broadway	153	
The Forest of Love	151	
Citation	151	
The Last Days of American Crime	149	
Raat Akeli Hai	149	
Ludo	149	
Army of the Dead	148	
Drive	147	

5. Retrieve the titles of Netflix Originals along with their respective genres.

```
SELECT
    no.title,
    gd.genre_name
FROM netflix_originals AS no
INNER JOIN genre_details AS gd
    ON no.genre_id = gd.genre_id
ORDER BY gd.genre_name, no.title;
```

Output	Result #1	
title	genre_name	
#REALITYHIGH	Action	
13th	Action	
13th: A Conversation with Oprah Winfrey & Ava DuVernay	Action	
15-Aug	Action	
1922	Action	
22-Jul	Action	
5 Star Christmas	Action	
6 Balloons	Action	
6 Underground	Action	
7 años	Action	

6. Rank Netflix Originals based on their IMDb scores within each genre

```
SELECT
    gd.genre_name,
    no.title,
    no.imdb_score,
    RANK() OVER (
        PARTITION BY no.genre_id
        ORDER BY no.imdb_score DESC
    ) AS imdb_rank
FROM netflix_originals AS no
INNER JOIN genre_details AS gd
    ON no.genre_id = gd.genre_id
ORDER BY gd.genre_name, imdb_rank;
```

Output	Result #1			
genre_name	title	imdb_score	imdb_rank	
Action	The Mars Generation	6.4	5017	
Action	Irreplaceable You	6.4	5017	
Action	The Heartbreak Club	6.4	5017	
Action	Sturgill Simpson Presents: Sound & Fury	6.4	5017	
Action	The Willoughbys	6.4	5017	
Action	Isi & Ossi	6.4	5017	
Action	Take Your Pills	6.4	5017	
Action	My Own Man	6.4	5017	
Action	Over the Moon	6.4	5017	
Action	Nappily Ever After	6.4	5017	
Action	Layla Majnun	6.4	5017	
Action	Street Flow	6.4	5017	
Action	Strong Island	6.4	5017	

7. Which Netflix Originals have IMDb scores higher than the average IMDb score of all titles?

```
SELECT
    title,
    imdb_score
FROM netflix_originals
WHERE imdb_score > (
    SELECT AVG(imdb_score)
    FROM netflix_originals
)
ORDER BY imdb_score DESC;
```

Output	Result #1
title	imdb_score
David Attenborough: A Life on Our Planet	9.0
Emicida: AmarElo - It's All For Yesterday	8.6
Springsteen on Broadway	8.5
Ben Platt: Live from Radio City Music Hall	8.4
Taylor Swift: Reputation Stadium Tour	8.4
Winter on Fire: Ukraine's Fight for Freedom	8.4
Cuba and the Cameraman	8.3
Dancing with the Birds	8.3
13th	8.2
Disclosure: Trans Lives on Screen	8.2
Klaus	8.2
Seaspiracy	8.2

8. How many Netflix Originals are there in each genre?

```
SELECT
    gd.genre_name,
    COUNT(no.title) AS total_netflix_originals
```

```

FROM netflix_originals AS no
INNER JOIN genre_details AS gd
    ON no.genre_id = gd.genre_id
GROUP BY gd.genre_name
ORDER BY total_netflix_originals DESC;

```

genre_name	total_netflix_originals
Horror	584
Musical	584
Drama	584
Comedy	584
Action	584
Mystery	584
Science Fiction	584
Thriller	584
Satire	584
Variety Show	584
Rom-Com	584
Concert Film	584
Biopic	584
Historical	584

9. Which genres have more than 5 Netflix Originals with an IMDb score higher than 8?

```

SELECT
    gd.genre_name,
    COUNT(no.title) AS titles_above_8
FROM netflix_originals AS no
INNER JOIN genre_details AS gd
    ON no.genre_id = gd.genre_id
WHERE no.imdb_score > 8
GROUP BY gd.genre_name

```

HAVING COUNT(no.title) > 5

ORDER BY titles_above_8 DESC;

Output	Result #1
genre_name	titles_above_8
Horror	16
Musical	16
Drama	16
Comedy	16
Action	16
Mystery	16
Science Fiction	16
Thriller	16
Satire	16
Variety Show	16

10. What are the top 3 genres with the highest average IMDb scores, and how many Netflix Originals do they have?

SELECT

gd.genre_name,

ROUND(AVG(no.imdb_score), 2) AS average_imdb_score,

COUNT(no.title) AS total_netflix_originals

FROM netflix_originals AS no

INNER JOIN genre_details AS gd

ON no.genre_id = gd.genre_id

GROUP BY gd.genre_name

ORDER BY average_imdb_score DESC

LIMIT 3;

Output	Result #1	
genre_name	average_imdb_score	total_netflix_originals
Horror	6.27	584
Musical	6.27	584
Drama	6.27	584